

Topics to be covered

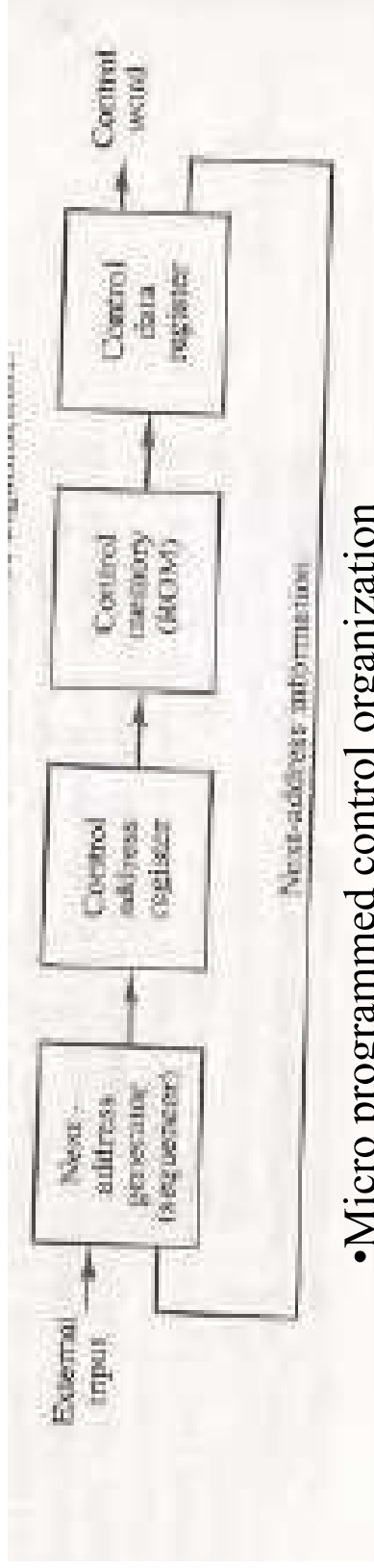
- Control Memory,
- Address sequencing,
- Micro program example,
- Design of Control Unit

Control Memory

- The function of the control unit in a digital computer is to initiate sequences of micro operations.
- The control unit initiates a series of sequential steps of micro operations.
- The control variables at any given time can be represented by a string of 1's and 0's called a *control word*.
- As such, control words can be programmed to perform various operations on the components of the system.
- A control unit whose binary control variables are stored in memory is called **micro programmed control unit**.
- A memory that is a part of control unit is called a control memory.

Address sequencing

- Control memory address register specifies the address of the microinstruction, and the control data register holds the microinstruction read from memory.
- The microinstruction contains a control word that specifies one or more micro operations for the data processor.
- The next address generator is sometimes called a micro program sequencer, as it determines the address sequence that is read from control memory.
- The control data register holds the present microinstruction while the next address is computed and read from memory. The data register is sometimes called a pipeline register.
- It allows the execution of the micro operations specified by the control word simultaneously with the generation of the next microinstruction.

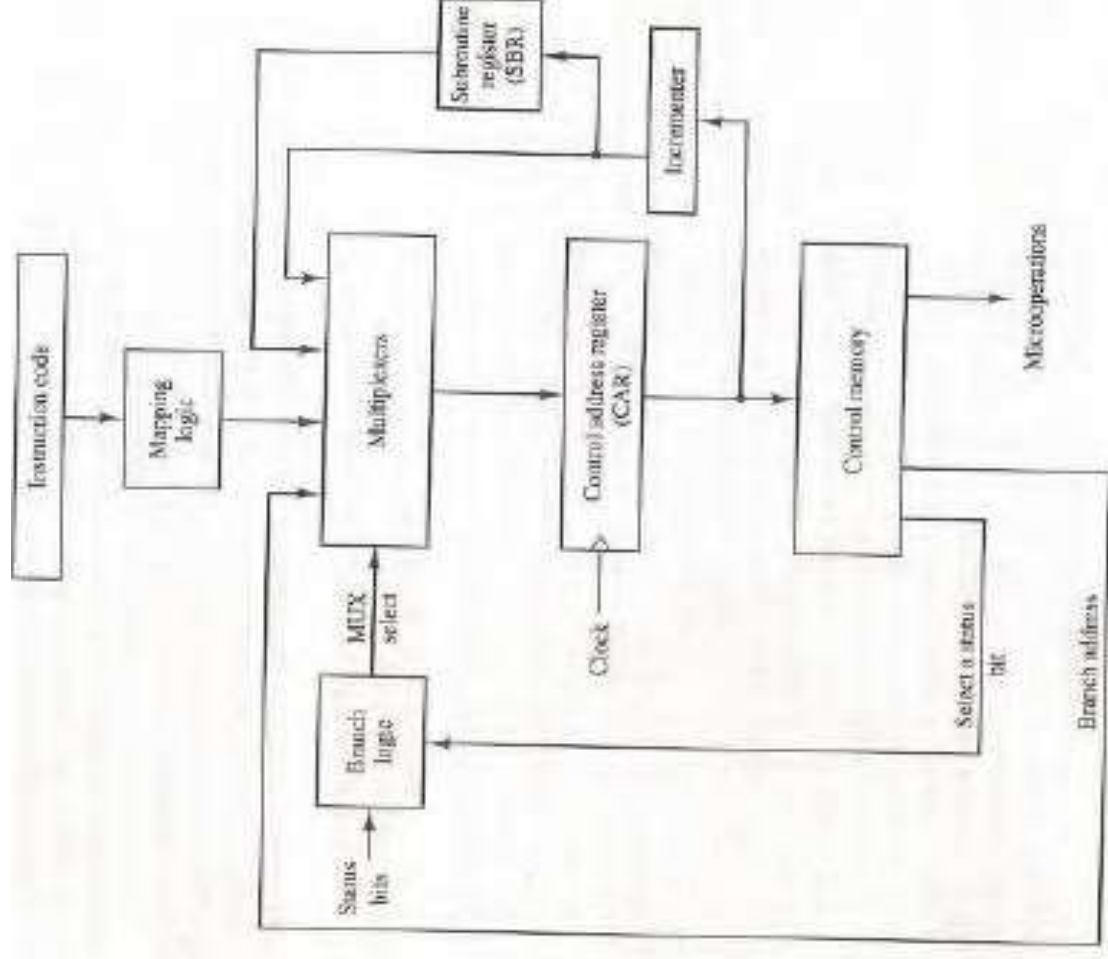


- Micro programmed control organization

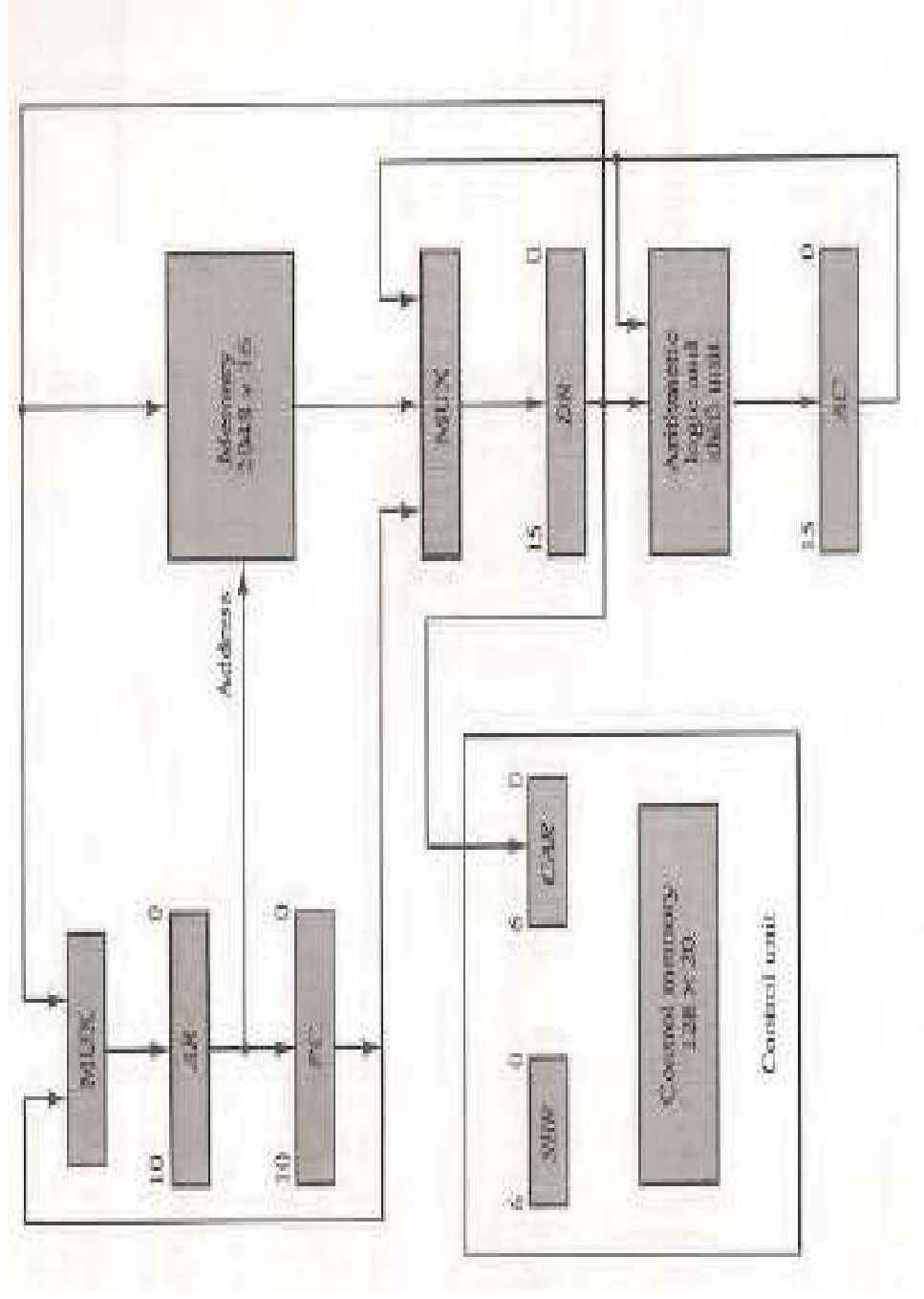
Address sequencing

The address sequencing capabilities required in a control memory are:

1. Incrementing of the control address register.
2. Unconditional branch or conditional branch, depending on status bit conditions.
3. A mapping process from the bits of the instruction to an address for control memory.
4. A facility for subroutine call and return.

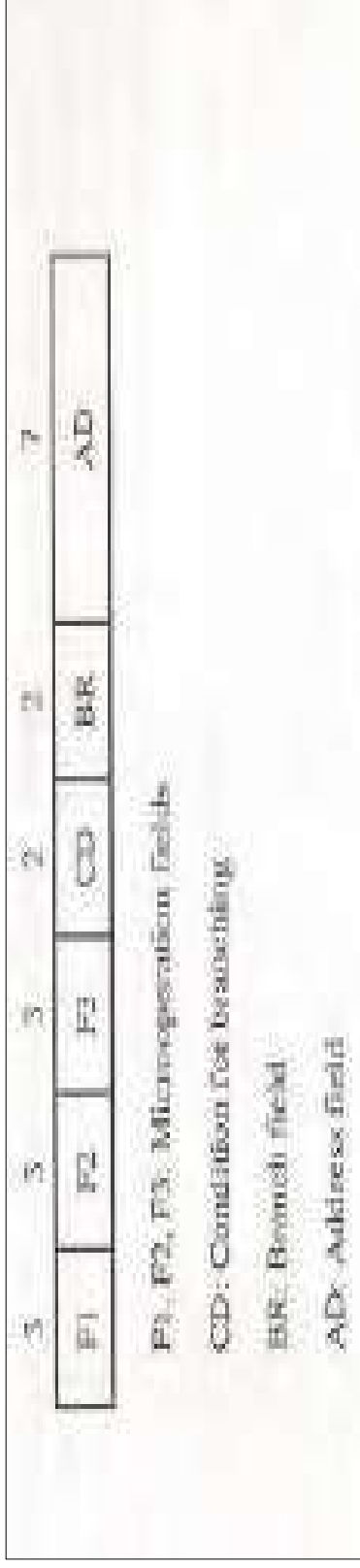


Micro program example



Computer hardware configuration.

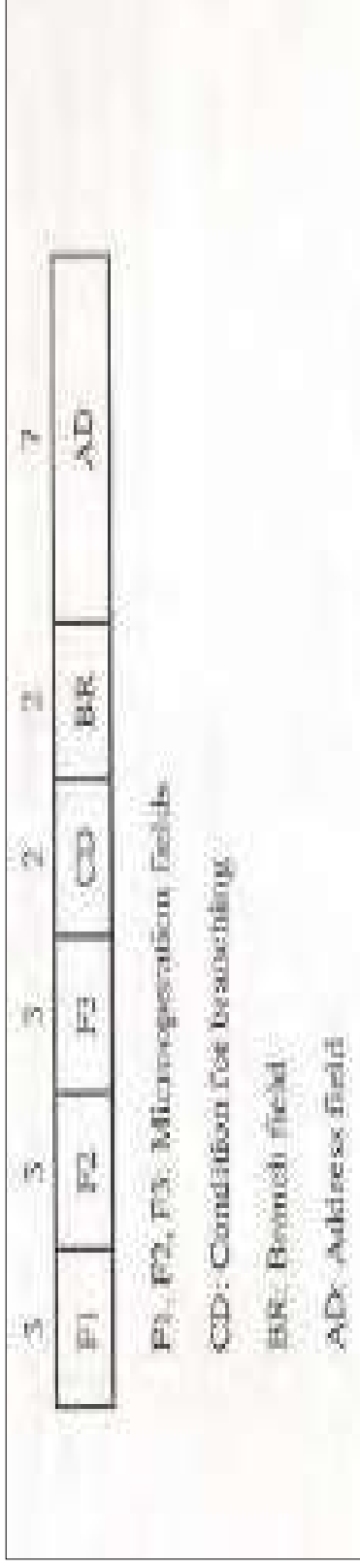
Micro program example



- Microinstruction code format (20 bits).

- The micro operations are subdivided into three fields of three bits each.
- The three bits in each field are encoded to specify seven distinct micro operations
- The CD (condition) field consists of two bits which are encoded to specify four status bit conditions
- The BR (branch) field consists of two bits. It is used, in conjunction with the address field AD

Micro program example



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Micro program example

The fields specify the following information.

1. The label field may be empty or it may specify a symbolic address. A label is terminated with a colon (:).
2. The microoperations field consists of one, two, or three symbols, separated by commas, . There may be no more than one symbol from each F field. The NOP symbol is used when the microinstruction has no micro operations. This will be translated by the assembler to nine zeros.
3. The CD field has one of the letters U, I, S, or Z.
4. The BR field contains one of the four symbols defined in previous Table
5. The AD field specifies a value for the address field of the microinstruction in one of three possible ways:
 - a. With a symbolic address, which must also appear as a label
 - b. With the symbol NEXT to designate the next address in sequence
 - c. When the RR field contains a RFT or MAP symbol the AD field is — — — — —

Micro program example

F1		Microoperation	Symbol
000		None	NOP
001		$AC \leftarrow AC + DR$	ADID
010		$AC \leftarrow 0$	CLRAC
011		$AC \leftarrow AC + 1$	INCAC
100		$AC \leftarrow DR$	DRTAC
101		$AR \leftarrow DR(0-10)$	DRTAR
110		$AR \leftarrow PC$	PCTAR
111		$M[AR] \leftarrow DR$	WRITE

F2		Microoperation	Symbol
000		None	NOP
001		$AC \leftarrow AC - DR$	SUB
010		$AC \leftarrow AC \vee DR$	OR
011		$AC \leftarrow AC \wedge DR$	AND
100		$DR \leftarrow M[AR]$	READ
101		$DR \leftarrow AC$	ACTDR
110		$DR \leftarrow DR + 1$	INCDR
111		$DR(0-10) \leftarrow PC$	PCDOR

F3		Microoperation	Symbol
000		None	NOP
001		$AC \leftarrow AC \oplus DR$	XOR
010		$AC \leftarrow AC$	COM
011		$AC \leftarrow \text{shl } AC$	SHL
100		$AC \leftarrow \text{shr } AC$	SHR
101		$PC \leftarrow PC + 1$	INCPC
110		$PC \leftarrow AR$	ARTPC
111		Reserved	

CD	Condition	Symbol	Comments
00	Always = 1	U	Unconditional branch
01	$DR(15)$	I	Indirect address bit
10	$AC(15)$	S	Sign bit of AC
11	$AC = 0$	Z	Zero value in AC

BR	Symbol	Function
00	JMP	$CAR \leftarrow AD$ if condition = 1 $CAR \leftarrow CAR + 1$ if condition = 0
01	CALL	$CAR \leftarrow AD$, $SBR \leftarrow CAR + 1$ if condition = 1 $CAR \leftarrow CAR + 1$ if condition = 0
10	RET	$CAR \leftarrow SBR$ (Return from subroutine)
11	HALT	End of program

Design of Control Unit

- The bits of the microinstruction are usually divided into fields, with each field defining a distinct, separate function.
- The various fields encountered in instruction formats provide control bits to initiate microoperations in the system, special bits to specify the way that the next address is to be evaluated, and an address field for branching.
- The number of control bits that initiate microoperations can be reduced by grouping mutually exclusive variables into fields and encoding the k bits in each field to provide 2^k micro operations

Design of Control Unit

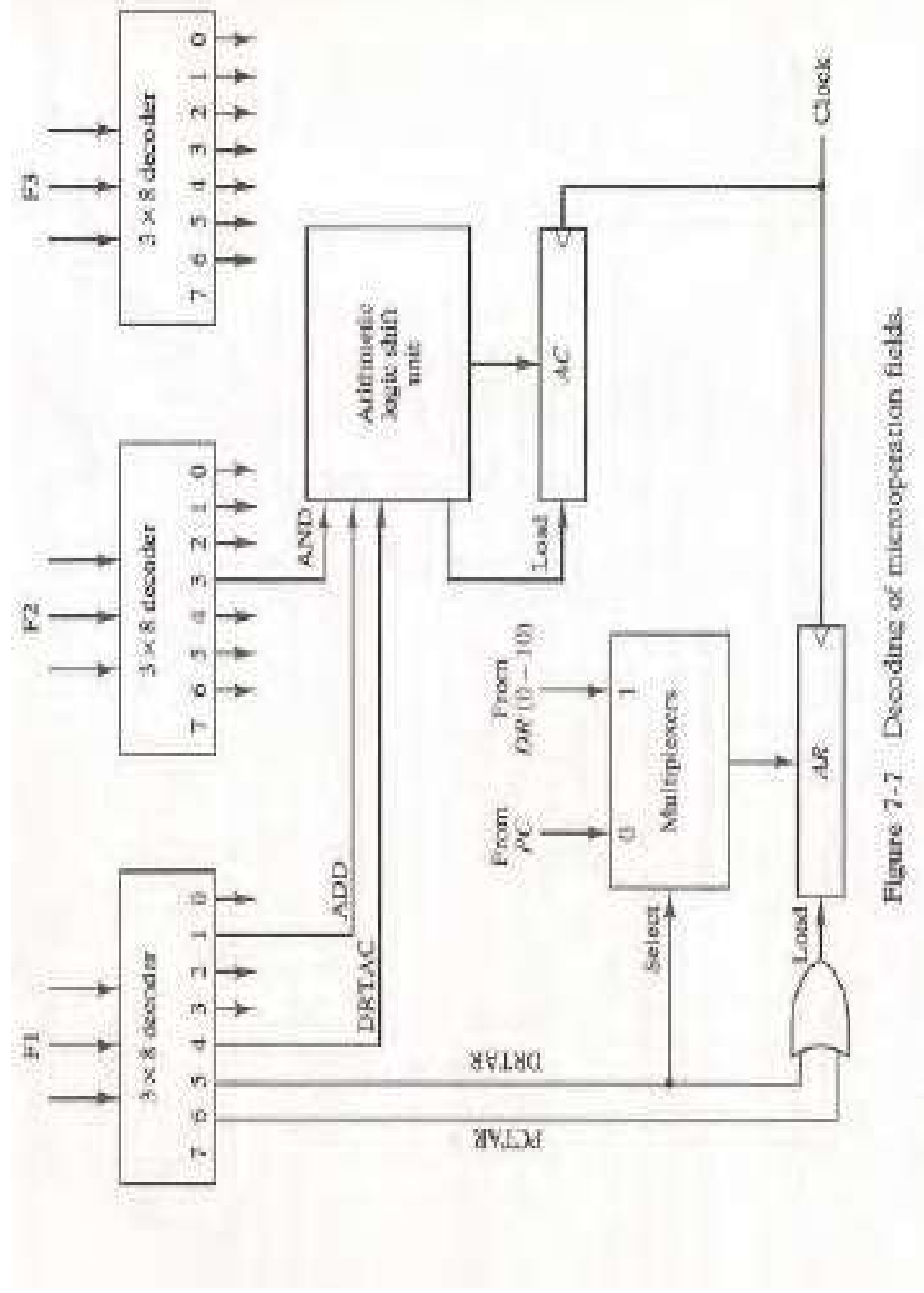


Figure 7-7 Decoding of microoperation fields.

Design of Control Unit

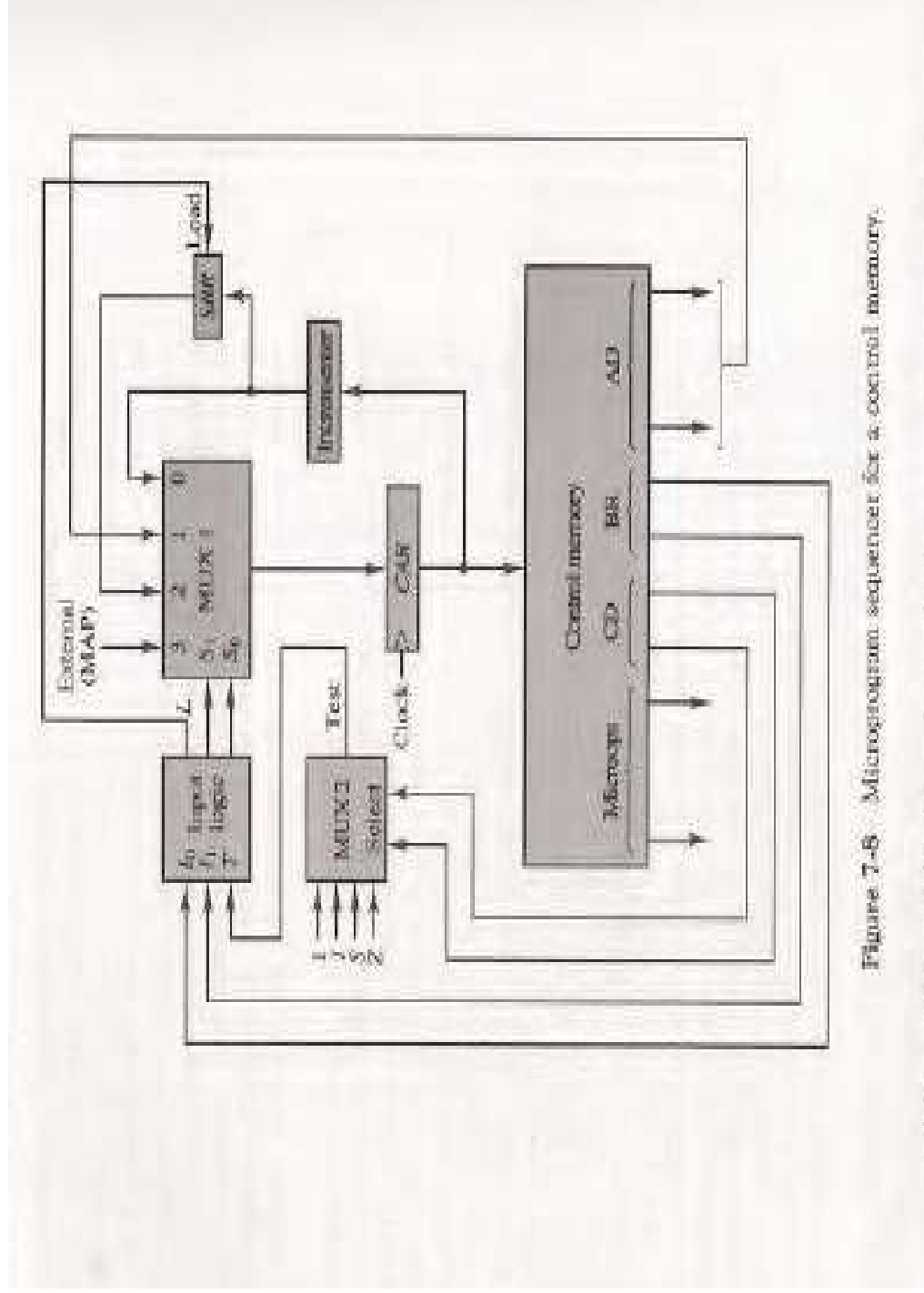


Figure 7-6 Microprogram sequencer for a control memory.

References

- Images , descriptive Tables , from Computer System Architecture, Morris Mano, 3rd edition Prentice Hall
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